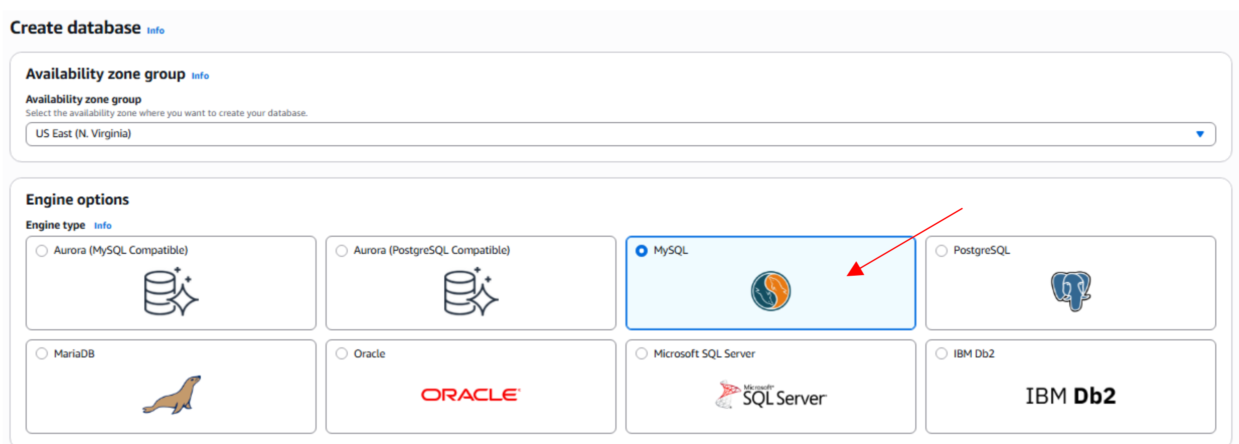
AWS RDS Connectivity

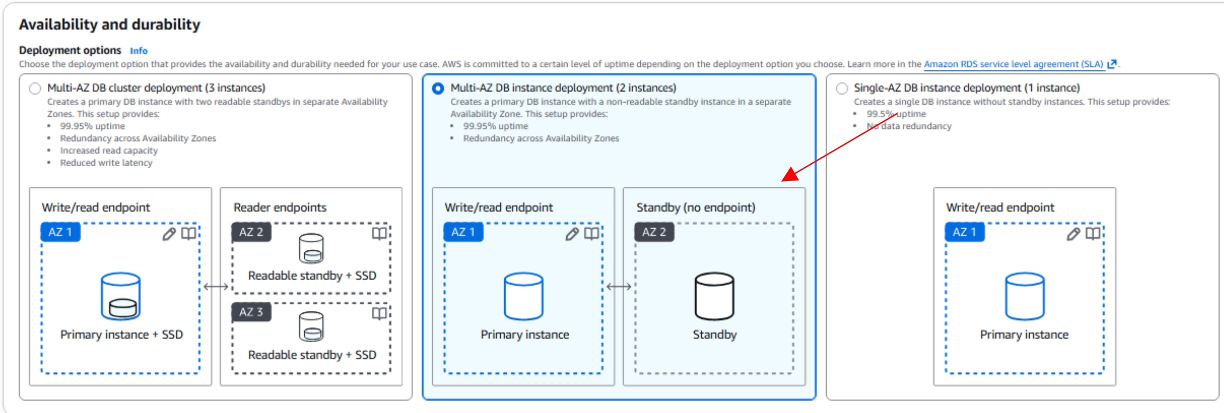
Summary: We use Amazon Web Services RDS (Relational Database Service) connectivity to allow applications, websites, or servers such as Amazon EC2 instances to securely connect with a managed database like MySQL, PostgreSQL, or Oracle. It helps applications store, retrieve, and manage data without manually handling database server maintenance, backups, updates, or scaling.

The need to modify RDS connectivity settings instead of using default configurations is mainly for:

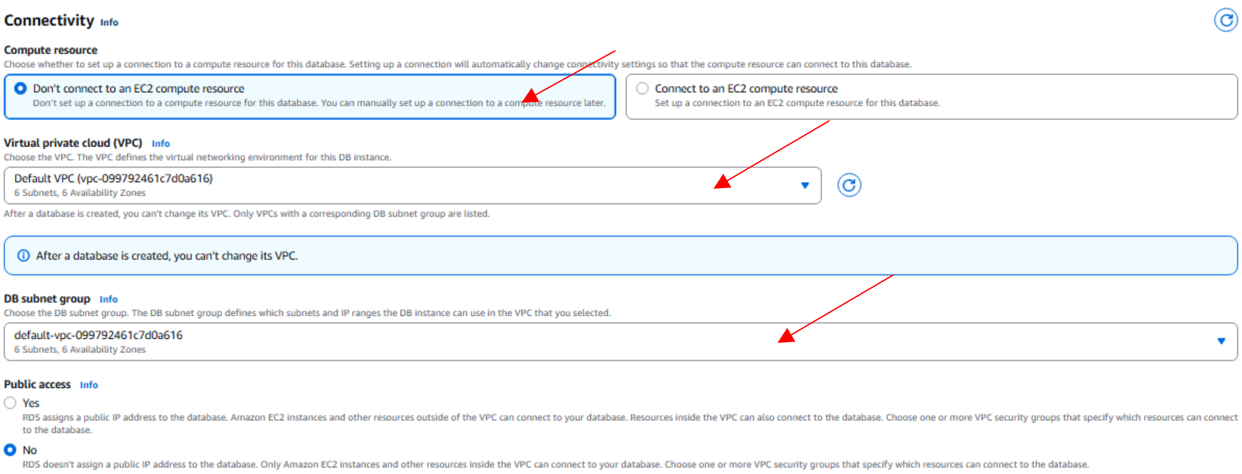
- **Security** – Default settings may not be secure for real projects. We modify security groups, ports, and public/private access to control who can access the database.
- **Performance** – Applications may require optimized storage, memory, or connection settings for faster response.
- **Scalability** – Default configurations may not handle high traffic, so settings are changed for better scaling and load handling.
- **Availability** – Features like Multi-AZ deployment and automated backups are configured to improve reliability.
- **Customization** – Different projects need different database engines, parameter groups, and networking setups.
- **Cost Optimization** – Default resources may be unnecessary or expensive, so configurations are adjusted based on project needs.



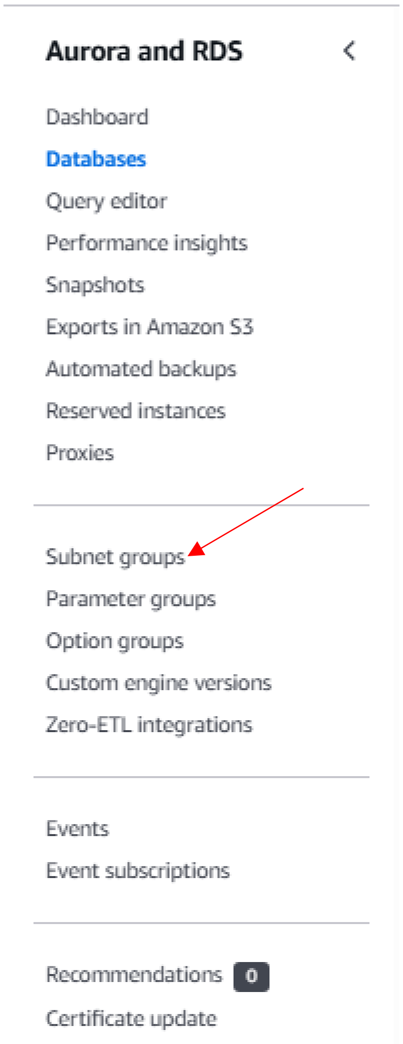
- To set up connectivity, we first create a database instance in RDS.



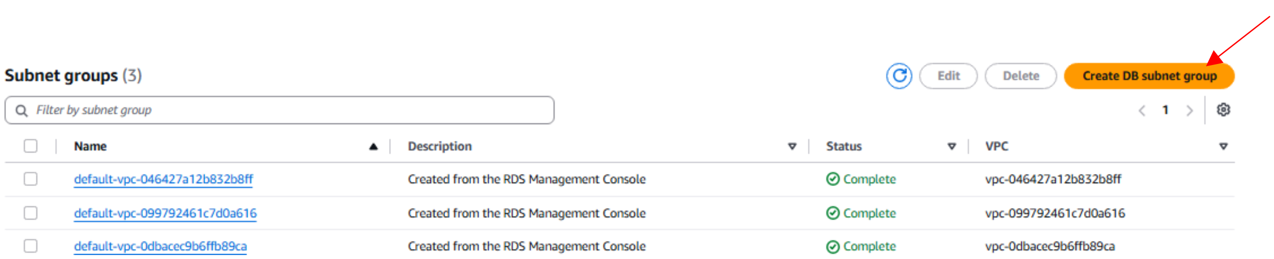
- For this guide, we are using the **Multi-AZ DB instance deployment (2 instances)** deployment option. This configuration automatically provisions and manages two database instances across two separate Availability Zones, providing high availability and automatic failover.



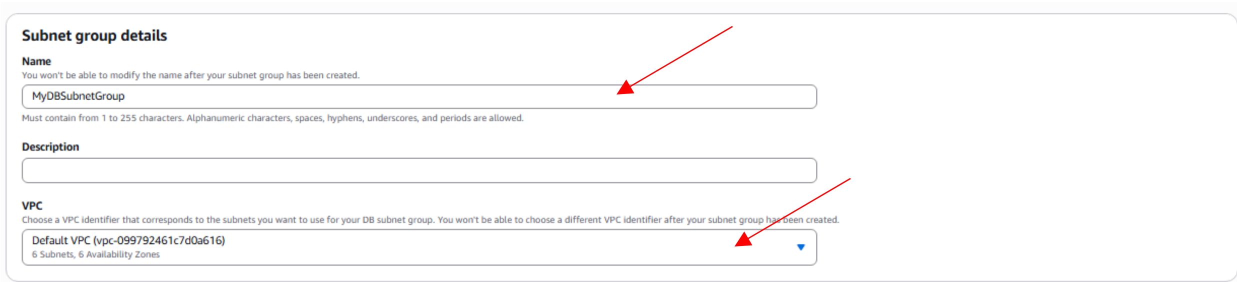
- Select the VPC (Virtual Private Cloud) and DB subnet group. The VPC defines the isolated network environment in which your database will reside, while the DB subnet group specifies which subnets within that VPC the database can use.
- Before creating the database, you must first create a DB subnet group. A DB subnet group is a collection of subnets (in different Availability Zones) that AWS uses to place your RDS instances.



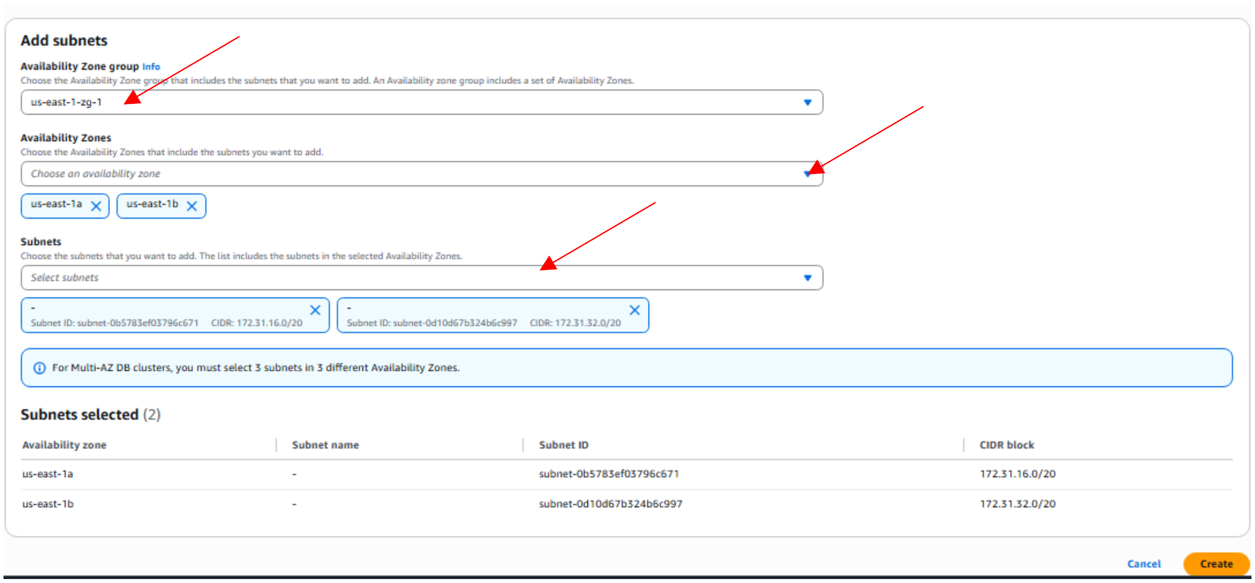
- In the RDS dashboard, navigate to the “Subnet groups” section and click “Create DB subnet group” to begin creating your subnet group.



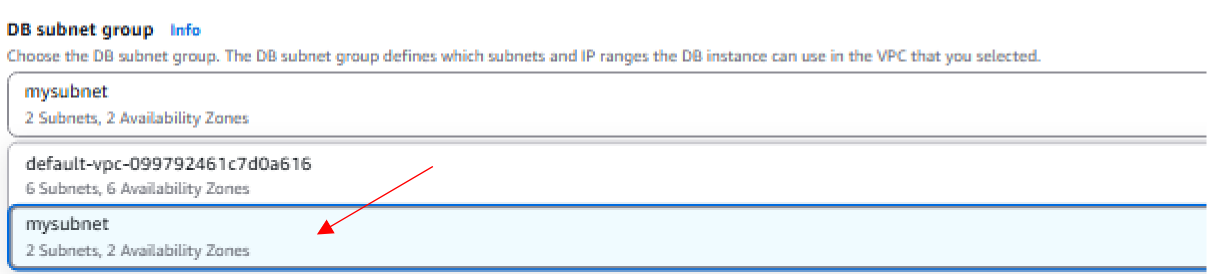
- AWS provides some default subnets, but for this setup we will create a new custom subnet group to have better control over networking.



- Enter a subnet name, add a description, and select the appropriate VPC from the dropdown list.

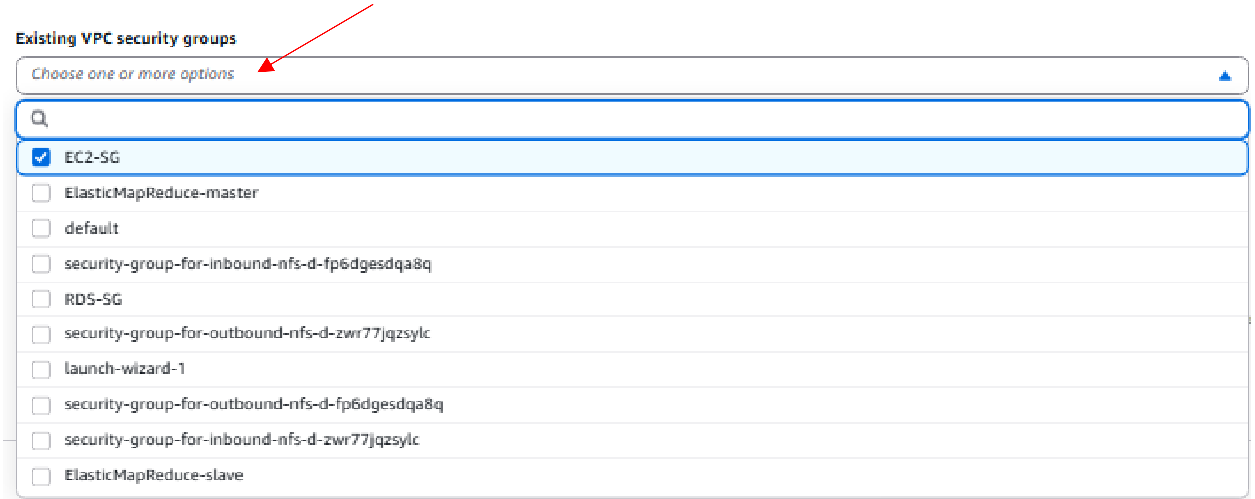


- Next, select the Availability Zones. Availability Zones are physically separate data centers within an AWS region. Placing your subnets in different zones ensures redundancy.
- Note: You must select at least 2 Availability Zones because we are creating a database using the **Multi-AZ DB instance deployment (2 instances). Each instance is placed in a different zone, which is required for failover support.**
- Select a specific subnet ID for each chosen Availability Zone. Each subnet ID corresponds to a defined IP address range within that zone.
- Click "Create" to finalize the DB subnet group. It will now be available for use when configuring your RDS database.

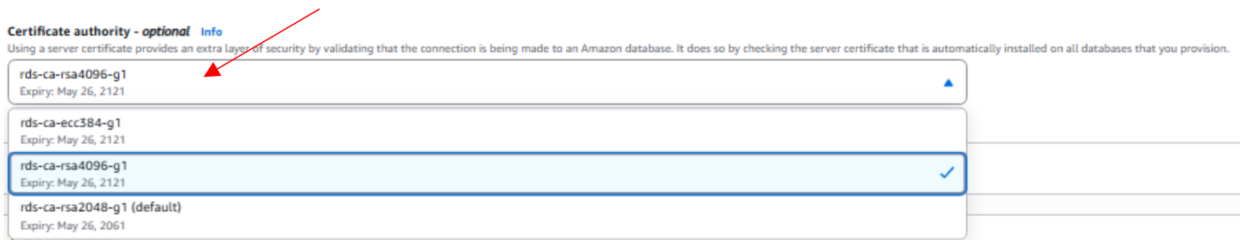


- Return to the Database creation page in the RDS console to continue with the remaining configuration steps.

- The subnet group you just created will now appear in the “DB subnet group” dropdown on the database creation page.
- Select the newly created subnet group from the list to associate it with your database.



- Set the **Existing VPC security groups** by selecting an existing VPC security group from the list. A security group acts as a virtual firewall, controlling which inbound and outbound traffic is allowed to reach your database. If you do not have one already, you can create a new security group through the VPC console before proceeding.



- Also select a **Certificate authority** to enable SSL/TLS encryption for data in transit. This ensures that all communication between your application and the RDS instance is encrypted, protecting sensitive data from interception.